Jatismar Saha

PhD Researcher | Lead-free Perovskite Photovoltaics (PV) | Thin Films | Semiconductor Device Simulation (TCAD)

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Google Scholar: scholar.google.com/citations?user=4rPTNDYAAAAJ

PROFILE

PhD researcher at IIT Guwahati working on lead-free, tin-based perovskite solar cells; currently a JST-LOTUS Fellow at the National Institute for Materials Science (NIMS), Japan. Hands-on perovskite solar cell fabrication in an inert-atmosphere glovebox, thermal evaporation and FESEM at NIMS, with XRD training in progress; cleanroom process training at IIT Guwahati in PECVD, RF sputtering and oxidation / diffusion furnace. Four peer-reviewed 2025 journal papers (three first-author) on tunnel FET simulation in Synopsys Sentaurus TCAD, including heavy-ion radiation effects; 11 citations, h-index 2 (Google Scholar). Open to research collaborations.

SKILLS

Photovoltaics and thin films	Perovskite solar cell fabrication, lead-free (tin-based) perovskites, photovoltaics (PV), inert-atmosphere glovebox operation, spin coating, thermal evaporation, RF sputtering, plasma-enhanced chemical vapor deposition (PECVD), oxidation / diffusion furnace, cleanroom processing
Materials characterization	Field-emission scanning electron microscopy (FESEM), X-ray diffraction (XRD, training in progress), UV-Visible spectroscopy (UV-Vis)
Device simulation	Synopsys Sentaurus TCAD, semiconductor device modeling, tunnel field-effect transistors (TFET), silicon-on-insulator (SOI), radiation effects (heavy-ion), interface trap charges
Design and programming	Python, Cadence Virtuoso, Xilinx Vivado, KiCad, Eagle PCB, Arduino

RESEARCH EXPERIENCE

JST-LOTUS Fellow May 2026 – Present

Photovoltaic Materials Group, Research Center for Energy and Environmental Materials (GREEN), National Institute for Materials Science (NIMS), Tsukuba, Japan · Host: Group Leader Yasuhiro Shirai

- Fabricate perovskite solar cells in an inert-atmosphere glovebox and deposit device layers by thermal evaporation.
- Characterize film morphology by FESEM; in XRD training for crystal structure and phase analysis.

Doctoral Researcher Jul 2025 – Present

School of Energy Science and Engineering, Indian Institute of Technology (IIT) Guwahati, India · Supervisor: Dr. Ajay Singh

- PhD research on lead-free, tin-based perovskite solar cells: fabrication, characterization and device modeling.
- Cleanroom process training: oxidation / diffusion furnace, PECVD, RF sputtering.

MTech Research, Semiconductor Device Simulation Aug 2023 – May 2025

National Institute of Technology (NIT) Silchar, India

- Simulated n-type step tunneling path TFETs and silicon-on-insulator (SOI) n-p-n double-gate TFETs in Synopsys Sentaurus TCAD.
- Simulated the effects of heavy-ion strikes, interface trap charges and temperature on TFET performance.
- Published 4 peer-reviewed journal papers in 2025, 3 as first author (Elsevier, Springer, Wiley, KIEEME/Springer).

PUBLICATIONS

PEER-REVIEWED JOURNAL ARTICLES

1. **J. Saha**, B. Choudhuri, R. Saha, "Radiation effects in silicon-on-insulator n-p-n double gate tunnel field effect transistor," *physica status solidi (a)*, vol. 222, no. 24, e202500615, 2025. doi: 10.1002/pssa.202500615
2. **J. Saha**, M. P. Gogoi, S. Tiwari, B. Choudhuri, R. Saha, "Investigating the effects of heavy-ion radiation on n-type step tunnelling path TFET," *Micro and Nanostructures*, vol. 207, 208269, 2025. doi: 10.1016/j.micrna.2025.208269
3. **J. Saha**, M. P. Gogoi, B. Choudhuri, R. Saha, "Investigating the effects of interface trap charges and temperature on n-type step tunneling path TFET," *Analog Integrated Circuits and Signal Processing*, vol. 125, no. 1, 17, 2025. doi: 10.1007/s10470-025-02505-x
4. A. Singh, P. Hazarika, **J. Saha**, M. P. Gogoi, R. Goswami, R. Saha, "Work function engineering on electrical parameters for step tunneling path TFET," *Transactions on Electrical and Electronic Materials*, vol. 26, no. 6, pp. 814–821, 2025. doi: 10.1007/s42341-025-00627-7

BOOK CHAPTER

5. **J. Saha**, R. Choudhuri, D. Kalita, J. Swarnakar, "Development of a smart railway platform system," in *Futuristic Trends in Network & Communication Technologies*, IIP Series, vol. 3, pp. 186–196, 2023. doi: 10.58532/V3BINC3P6CH5

HONORS AND FELLOWSHIPS

JST-LOTUS Fellowship, Japan Science and Technology Agency (JST) — 2026

EDUCATION

PhD, Energy Science and Engineering — Jul 2025 – Present
Indian Institute of Technology (IIT) Guwahati, India · Supervisor: Dr. Ajay Singh

MTech, Microelectronics and VLSI Design — Aug 2023 – May 2025
National Institute of Technology (NIT) Silchar, India · CGPA 7.48/10

BTech, Electronics and Communication Engineering — Aug 2019 – Jul 2023
North-Eastern Hill University (NEHU), Shillong, India · CGPA 7.25/10

INTERNSHIPS

PCB design for an indoor drone, Indian Institute of Technology (IIT) Guwahati · schematic and layout in KiCad and Eagle PCB — Oct 2024 – Jun 2025

Fixed-wing single-rotor aircraft development, IIT Guwahati — 2023

Hexacopter drone development, IIT Guwahati · Mission Planner — Aug – Sep 2022

PCB fabrication and 3D printing, IIT Guwahati — May – Jun 2022

LANGUAGES

English, Hindi, Assamese, Bengali; Japanese (beginner, currently taking classes)

REFERENCES

Contact details available on request.

Dr. Ajay Singh
Assistant Professor, School of Energy Science and Engineering, IIT Guwahati, India

Prof. Sisir K. Nayak
Professor, Department of Electronics and Electrical Engineering, IIT Guwahati, India

Prof. Vinayak Kulkarni
Professor, Department of Mechanical Engineering, IIT Guwahati, India

Dr. Rajesh Saha
Assistant Professor, Department of Electronics and Communication Engineering, NIT Silchar, India